Zeyad Manaa

Graduate Research and Teaching Assistant

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An Aerospace Engineering graduate research and teaching assistant who seeks a research intern applying data-driven methods and machine learning algorithms to problems in dynamical systems, and control.

Research Interests

- Machine learning and data science for modeling and control of dynamical systems
- Control theory, optimization, and AI for dynamical systems
- Model reduction, and feedback control of dynamical systems

Education

2023–present **M.Sc in Aerospace Engineering**, *King Fahd University for Petroleum & Minerals*, Dhahran, Saudi Arabia, Funded by the Interdisciplinary Research Center for Aviation & Space Exploration.

Thesis Data-driven Approaches for Dynamics Modeling and Control for Aerospace Applications – On Linear and Nonlinear Interpretable Machine Learning Techniques for Dynamics Identification and Control (Tentative Title)

Note Funded by the Interdisciplinary Research Center for Aviation & Space Exploration

2017–2022 **B.Sc in Aerospace Engineering**, *University of Science and Technology at Zewail City*, Giza, Egypt.

Thesis Development of the software package for the attitude determination and control algorithm of a cube satellite

Patents and Publications[1]

2024 [C1] **Zeyad M. Manaa**, Mohammed R Elbalshy, and Ayman M Abdallah. Data-driven discovery of the quadrotor equations of motion via sparse identification of nonlinear dynamics. In *AIAA SCITECH 2024 Forum*. AIAA, 2024.

2024 [J4] **Zeyad M. Manaa** and Naef Qassem. Novel airfoil for improved supersonic performance. *In Review in Aerospace Science and Technology*. Elsevier, 2024.

2024 [J3] **Zeyad M. Manaa**, Ayman M. Abdallah, Syed Saad Azhar Ali, and Mohammed Abido. Local koopman observables for identifying globally linear model for quadrotor's dynamics. *In Prep*, 2024.

2024 [J2] **Zeyad M. Manaa**, Ayman M. Abdallah, Syed Saad Azhar Ali, and Mohammed Abido. Identification of a globally linear model for quadrotor's dynamics using data-driven koopman theory. *In Prep*, 2024.

2024 [J1] **Zeyad M. Manaa** and Ayman M. Abdallah. Robust data-driven identification of quadrotor's equation of motion. *In Prep*, 2024.

2024 [P1] Naef Qassem and **Zeyad M. Manaa**. Efficient airfoil for improved supersonic performance for fighters. *Under Review*, 2024.

Research Experience

January 2023 **Research Assistant - Flight Dynamics & Control and UAV Lab**.
King Fahd University for Petroleum and Minerals, Saudi Arabia

Advisor Ayman Abdalla, *Chairman, Department of Aerospace Engineering, KFUPM*

- Research on **Koopman** operator modeling to globally linearize nonlinear dynamics and its intersection with deep learning.
- Worked on the sparse identification of nonlinear dynamics (**SINDy**) for complex dynamical systems (quadrotors).

June 2022 - August 2022 **Research Intern - Aerial Robotics Lab**.
National University of Science and Technology, Pakistan

Advisors Muhammad Jawad Khan and Muhammad Aleem Siddiqui

- Utilized hand gestures to control drones via built-in/web camera using **Computer Vision/Machine Learning**.
- Developed an autonomous control system for a quad-copter drone for Tello and COEX clover drones with **ROS**.

August 2021 - July 2022 **Research Intern - ADCS Lab**.
Egyptian Space Agency – University of Science and Technology at Zewail City, Egypt

Advisor Mustafa Abdalla, *Former Chairman, Department of Aerospace Engineering, Zewail City*

- Conducted the bachelor's thesis research under co-supervision of the Egyptian Space Agency and University of Science and Technology at Zewail City on spacecraft attitude determination and control subsystem.
- Developed the software of the **attitude determination and control** algorithm of a cube satellite.
- Studied and implemented the space environment as a means of Earth's Magnetic Field (IGRF Model), Earth's gravitational Field (Spherical Harmonics) as well as modeling the space disturbances using **MATLAB**.

Presentations & Talks

March 2024 **KIKX Seminar: Data-driven techniques for modeling and control in aerospace applications**, *Aerospace Dept. KFUPM*.

[1] [J#] is a keyword for journals, [C#] for conferneces, and [P#] for patents.

December 2023 **Presentation: Observations are all you need! - Koopman's theory unfolds globally linear model for quadrotor**, *Aerospace Dept. KFUPM.*

May 2023 **Presentation: On proper orthogonal decomposition and dynamic mode decomposition for aerodynamic applications**, Math. Dept. KFUPM.

May 2023 **Seminar: Sparse identification of nonlinear dynamics for quadrotor's equation of motion discovery**, *Aerospace Dept. KFUPM.*

March 2023 **Presentation: Convex optimization for supersonic airfoil design using linear flow theory**, *Aerospace Dept. KFUPM.*

Teaching

Spring, 2024 **AE 540 - Advanced Flight Mechanics**, *Graduate Course*, Aerospace Dept., KFUPM, **Teaching Assistant**.

Fall, 2023 **AE 426 - Introduction to Flight Mechanics**, *Undergrad. Course*, Aerospace Dept., KFUPM, **Instructor**.

Fall, 2023 **AE 315 - Systems and Control**, *Undergrad. Lab*, Aerospace Dept., KFUPM, **Instructor**, (Link).

Fellowships & Awards

2023 **Graduate International Research Assistance Scholarship**, *King Fahd University for Petroleum and Minerals, Saudi Arabia.*

- Awarded a fully funded scholarship to study M.Sc degree in aerospace engineering for 2.5 years.

2023 **Mohammad Al-Aqeel Grant for Graduate Students**, *King Fahd University for Petroleum and Minerals, Deanship of Research Oversight and Coordination, Saudi Arabia.*

- Excellence award which is awarded to graduate students based on their performance and research outcomes.
- Granted to attend ***AIAA, SCITECH*** conference in Orlando, USA, 2024 to present [C1] paper.

2022 **Fully Funded Internship Program for International students**, *National University of Science and Technology, Pakistan.*

- A research intern scholarship at Ariel Robotics Lab at National University of Science and Technology.

2022 **Future Work is Digital Scholarship**, *Ministry of Communications and Information Technology, Egypt.*

- A fully funded scholarship given to students to study Data Science.

2019 **Smart City Hackathon**, *Digital Arabia Network, and Global Project Partners, Germany.*

- Global finalist, awarded the **1st place** award.

2017 **Full Undergraduate Scholarship**, *University of Science and Technology at Zewail City, Egypt.*

- Merit-based sholarship given to high school student from the University of Science and Technology at Zewail City to cover five years of study.

Skills

Programming Languages	Python, MATLAB, C++, TeX, and basic ability with Julia
Machine Learning	Pytorch, TensorFlow, OpenCV, Sci-Kit
Other Skills	GitHub, Mathematica, bash-scripting, and LabVIEW
Platforms	Linux, Windows, and MacOS
Relevant Coursework	Machine Learning, Advanced Dynamics, Flight Mechanics, Linear Algebra, Discrete Math, Probability and Statistics, ODE, PDE, Modern Control, Intelligent Control, Communication Theory
Professional Memberships	AIAA student member
Languages	Arabic (Mother Tongue), English (C1 user, IELTS: 7)

Professional Service

Reviewer for Conference Proceedings

2023 **AIAA, SCITECH 2024 Conference**.

Leadership

2020 **Euroavia Egypt**.

Managed a team of 10 people as a the Media committee head for the Euroavia Egypt, student branch

2019 **Zewail City Science Festival**.

Managed a team of 15 people as a the Media committee head for the Zewail City Science Festival mega even, Egypt.

Referees

Available upon request

Last Updated: January 17, 2024